Demand and Welfare

Session 3

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Where household surveys earn their keep for policy: what happens to whom when a price moves.

Reading

- Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)
- Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326
- Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
- Ligon, "Household Welfare from Disaggregate Demands" (working paper) — the CFE system we estimate today

Budget shares and Engel curves

Let $w_{ij} = p_j q_{ij}/x_i$ be household i's budget share on good j, x_i total expenditure. The Working–Leser form,

$$w_{ij} = \alpha_j + \beta_j \log x_i + \gamma_j' z_i + u_{ij},$$

fits food shares remarkably well across countries and decades.

- $\beta_{\text{food}} < 0$ is Engel's law
- z_i: household size and composition, region, season
- Estimated by OLS, weighted, clustered on the EA

Equivalence scales from behaviour

Session 2 *assumed* a scale. Demand analysis lets you estimate one.

- Engel method: two households are equally well off if they have the same food share. Scale = the expenditure ratio that equalizes it.
- Rothbarth method: use expenditure on adult goods (alcohol, tobacco, adult clothing) as the indicator instead.
- The two disagree, systematically and by a lot. Deaton ch. 4 explains why: the Engel method conflates the cost of a child with the child's effect on the household's needs for food specifically.

Prices, and where to get them

LSMS surveys rarely carry a price questionnaire. They carry unit values, and unit values are contaminated by quality choice:

$$\log v_{ij} = \log p_j + \underbrace{\psi_j \log x_i}_{\text{quality}} + \epsilon_{ij}.$$

Deaton's solution exploits the design: *within a cluster*, all households face the same price. So

- regress $\log v_{ij}$ on $\log x_i$ and household characteristics *with cluster fixed effects* to strip out quality;
- the cluster effects are the price variation you wanted.

This is the single best argument for caring about v in the index.

The rank of a demand system

Stack the Engel curves. The *rank* of a demand system (Lewbel 1991) is the dimension of the space spanned by the budget shares viewed as functions of total expenditure.

- rank 1: shares do not vary with x — homothetic
- rank 2: Working–Leser, AIDS
- rank 3: QUAIDS; Gorman's bound for exactly aggregable systems

Why you should care: if the rank exceeds one, no single price index can deflate nominal expenditure so as to keep it monotone in utility when relative prices move. Rank r needs r indices.

Constant Frisch Elasticity demands

Let λ_i be household i's marginal utility of expenditure. Frischian (λ-constant) demands take the CFE form

$$f_j(p, \lambda) = \left(\frac{\alpha_j}{\lambda p_j}\right)^{\beta_j}, \qquad U(c) = \sum_j \alpha_j \beta_j \frac{c_j^{1-1/\beta_j} - 1}{\beta_j - 1}.$$

- β_j is the Frisch elasticity of demand for good j: necessities small, luxuries large
- $\beta_j = \beta$ for all j gives CES; $\beta \to 1$ gives Cobb–Douglas
- unrestricted rank, so it does not impose what we just said is false

The estimating equation

Take logs of expenditure on good j. Writing $w_i \equiv -\log \lambda_i$,

$$\log x_{it}^j \; = \; a_t^j \; + \; \beta_j \, w_{it} \; + \; \gamma_j' d_{it} \; + \; \varepsilon_{it}^j.$$

This is a one-factor model. The $(it) \times j$ matrix of log expenditures, once good-time effects a_t^j and characteristics d_{it} are swept out, is rank one: loadings β, factor w.

- estimate by SVD of the residual matrix, with missing data
- d_{it} enters as good-specific Barten scales, not "independence of base"
- needs no prices and no total expenditure — only expenditures on a subset of goods

$w = -\log\lambda$ as a welfare measure

λ is the marginal utility of expenditure, so it *falls* as a household gets better off. Hence $w = -\log\lambda$ is increasing in welfare.

- comparable across households facing different prices — the prices are in a_t^j, not in w
- comparable across household compositions — composition is in $\gamma_j' d$
- derived from behaviour, not from an assumed equivalence scale or an assumed price index
- w is the partial derivative of indirect utility with respect to total expenditure: a genuine welfare object, though not itself a utility level

Welfare consequences of a price change

For a small change in the price of good j, the compensating variation is approximately

$$\mathrm{CV}_i \;\approx\; q_{ij}\,\Delta p_j \;=\; w_{ij} x_i \frac{\Delta p_j}{p_j}.$$

- to first order you need only the *budget share* — no elasticities
- the distributional incidence follows from how w_{ij} varies with x_i, which is the Engel curve you already estimated
- second-order terms need the demand system, and are usually small relative to the first-order term for modest price changes

Exercises

1. Estimate the Working–Leser food share equation on the *full* consumption aggregate rather than food alone, and test whether β differs between urban and rural households.
2. Implement the Engel equivalence scale: find the expenditure ratio that equates the predicted food share of a household with two adults and two children to that of a household with two adults. Compare with the θ-scale you used in session 2.
3. Compute the first-order welfare cost of a 20% rise in the price of the largest food item, by consumption decile. Who loses most, in cedis and as a share of expenditure? Are those the same households?
4. Re-estimate the CFE system using the enumeration area as the market m rather than a single national market. How much do the β_j move? What have you assumed away by using one market?